

COMPUTER FUNDAMENTALS

Unit-IV: Computer Software

Comprehensive Study Notes

1. Computer Software

1.1 What is Computer Software?

Computer software is a collection of **programs, data, and instructions** that tell a computer how to perform specific tasks. Software is the **non-physical part** of a computer system that makes the hardware functional.

1.2 Need for Software

Reason	Description
Communication	Software acts as a bridge between user and hardware
Task Execution	It enables computers to perform specific tasks
Resource Management	Software manages hardware resources efficiently
User Interface	Provides a user-friendly interface for interaction
Automation	Automates repetitive tasks for efficiency
Data Processing	Enables processing, storage, and retrieval of data
Connectivity	Allows networking and internet connectivity
Security	Provides protection against threats and unauthorized access

2. Types of Software

2.1 Classification of Software

```
Software
|
├─ System Software
|   ├─ Operating System (Windows, Linux, DOS)
|   ├─ Utility Programs (Antivirus, Disk Cleanup)
|   └─ Language Translators (Compiler, Interpreter, Assembler)
|
└─ Application Software
    ├─ Word Processing (MS Word)
    ├─ Spreadsheet (MS Excel)
    ├─ Presentation (MS PowerPoint)
    ├─ Database (MS Access)
    ├─ Graphics (Photoshop)
    └─ Browsers (Chrome, Firefox)
```

2.2 System Software

Definition

System software is a type of computer program designed to **manage and control** the computer hardware and provide a platform for running application software.

Types of System Software

Type	Description	Examples
Operating System	Manages hardware and software resources	Windows, Linux, macOS, DOS
Utility Programs	Performs maintenance and optimization tasks	Antivirus, Disk Cleanup, Backup

Programming Languages	Used to write computer programs	C, C++, Java, Python
Assemblers	Converts assembly language to machine code	MASM, NASM
Compilers	Converts high-level language to machine code	GCC, Turbo C++
Interpreters	Executes code line by line	Python Interpreter, BASIC

2.3 Application Software

Definition

Application software is a program designed to **perform specific tasks** for the user, such as word processing, calculations, or graphic design.

Types of Application Software

Type	Purpose	Examples
Word Processing	Create, edit, format documents	MS Word, Google Docs, Notepad
Spreadsheet	Organize, analyze, calculate data	MS Excel, Google Sheets
Presentation Graphics	Create visual presentations	MS PowerPoint, Prezi
Database	Store, manage, retrieve data	MS Access, Oracle, MySQL
Graphics	Create, edit images and designs	Photoshop, CorelDraw, GIMP
Desktop Publishing	Design printed materials	Adobe InDesign, Publisher

Multimedia	Create and edit media files	VLC, Adobe Premiere
Web Browser	Access and browse the internet	Chrome, Firefox, Edge
Email	Send and receive emails	Outlook, Gmail, Thunderbird
Antivirus	Protect against malware	Norton, McAfee, Kaspersky
Games	Entertainment	Minecraft, Call of Duty

3. Operating System

3.1 What is an Operating System?

An Operating System (OS) is the **most important system software** that acts as an interface between the **user and the computer hardware**. It manages all hardware resources and provides services to application software.

3.2 Functions of Operating System

Function	Description
Memory Management	Allocates and deallocates memory to processes
Process Management	Manages processes and CPU scheduling
File Management	Manages files, directories, and storage
Device Management	Controls input/output devices
Security	Provides protection against unauthorized access
User Interface	Provides command-line or graphical interface
Error Detection	Detects and handles errors
Resource Allocation	Allocates resources to processes

3.3 Types of Operating Systems

Type	Description	Examples
Batch OS	Processes jobs in batches	Early mainframe systems
Time-Sharing OS	Multiple users share CPU time	UNIX, Linux

Real-Time OS	Responds within strict time limits	VxWorks, QNX
Multi-User OS	Supports multiple users simultaneously	UNIX, Linux, Windows Server
Multi-Tasking OS	Runs multiple tasks simultaneously	Windows, Linux, macOS
Distributed OS	Manages network of computers	Windows Server, Linux Cluster
Embedded OS	Designed for embedded systems	Android, iOS, Raspberry Pi OS

4. Operating Systems for PCs

4.1 DOS (Disk Operating System)

Feature	Description
Full Form	Disk Operating System
Developer	Microsoft
Type	Single-user, single-tasking
Interface	Command-line (Text-based)
Release	1981
File System	FAT (File Allocation Table)
Popular Versions	PC-DOS, MS-DOS 6.22

DOS Features

Feature	Description
Command-Line Interface	User types commands to perform tasks
Single-User	Only one user at a time
Single-Tasking	One program at a time
No GUI	No graphical user interface
FAT Support	Uses FAT12, FAT16 file systems

Basic DOS Commands

Command	Function
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DIR	Lists files and directories
COPY	Copies files
DEL	Deletes files
FORMAT	Formats a disk
MD	Creates a directory
CD	Changes directory
RD	Removes directory
TYPE	Displays file content

4.2 Windows

Feature	Description
Developer	Microsoft
Type	Multi-user, multi-tasking
Interface	Graphical User Interface (GUI)
First Release	1985 (Windows 1.0)
Popular Versions	Windows 10, Windows 11, Windows Server

Windows Versions History

Version	Year	Key Features
Windows 1.0	1985	First GUI version
Windows 3.1	1992	Improved GUI
Windows 95	1995	Start Menu, Plug and Play
Windows XP	2001	Stable, user-friendly

Windows 7	2009	Improved interface
Windows 8	2012	Touch interface, Start Screen
Windows 10	2015	Universal apps, Cortana
Windows 11	2021	New UI, Android apps

Windows Features

Feature	Description
GUI	Graphical User Interface
Multi-Tasking	Multiple applications simultaneously
Plug and Play	Automatic hardware detection
File System	NTFS, FAT32
Security	Windows Defender, Firewall
Compatibility	Wide range of software and hardware
Network Support	Built-in networking features

4.3 Linux

Feature	Description
Developer	Linus Torvalds (Open Source)
Type	Multi-user, multi-tasking
Interface	GUI and Command-line
First Release	1991
Kernel	Linux Kernel
License	GNU GPL (Open Source)

Linux Distributions

Distribution	Description	Features
Ubuntu	Most popular Linux distro	User-friendly, Large community
Fedora	Cutting-edge features	Latest software
CentOS	Enterprise stability	RHEL compatible
Debian	Most stable	Free software only
Linux Mint	Easy to use	Windows-like interface
RHEL	Enterprise Linux	Commercial support
Kali Linux	Security focused	Penetration testing

Linux Features

Feature	Description
Open Source	Free to use and modify
Multi-User	Multiple users simultaneously
Multi-Tasking	Efficient process management
Security	High security, less viruses
Stability	Very stable for servers
Command Line	Powerful terminal interface
File System	ext3, ext4, XFS

5. File Allocation Table (FAT)

5.1 What is FAT?

FAT (File Allocation Table) is a **file system** developed by Microsoft that manages how files are stored and organized on a storage device.

5.2 Types of FAT

Type	Max Volume Size	Max File Size	Usage
FAT12	32 MB	32 MB	Floppy disks
FAT16	2 GB	2 GB	Older hard drives
FAT32	8 TB	4 GB	USB drives, older Windows

5.3 FAT32 Features

Feature	Description
Cluster Size	4 KB to 32 KB
Drive Size	Up to 8 TB
File Size Limit	4 GB maximum file size
Compatibility	Works with all OS
Simple	Easy to implement
Limitation	Cannot handle files larger than 4 GB

5.4 FAT32 vs NTFS

Feature	FAT32	NTFS
Max File Size	4 GB	16 TB
Max Partition Size	8 TB	256 TB
File Permissions	No	Yes
Security	No	Yes
Compression	No	Yes
Encryption	No	Yes
Compatibility	All OS	Windows only
Recovery	Limited	Advanced

6. Files and Directory Structure

6.1 File

What is a File?

A file is a **collection of related data** or information stored on a storage device with a specific name.

File Naming Rules

Rule	Description	Example
File Name	Maximum 255 characters	mydocument
Extension	3-4 characters after dot	.txt, .doc, .xls
Allowed Characters	Letters, numbers, and some symbols	file123.txt
Forbidden Characters	/ : * ? " < >	Cannot use these
Case Sensitive	Linux is case sensitive; Windows is not	File.txt vs file.txt
Spaces	Can use spaces	My File.doc

File Extensions

Extension	File Type	Program
.txt	Text file	Notepad
.doc/.docx	Word document	Microsoft Word
.xls/.xlsx	Excel spreadsheet	Microsoft Excel
.ppt/.pptx	PowerPoint presentation	Microsoft PowerPoint

.pdf	Portable Document Format	Adobe Reader
.jpg/.jpeg	Image file	Photo Viewer
.png	Image file	Photo Viewer
.mp3	Audio file	Media Player
.mp4	Video file	Media Player
.exe	Executable file	Program
.html	Web page	Browser
.zip	Compressed file	WinZip, WinRAR

6.2 Directory Structure

What is a Directory?

A directory (or folder) is a **container** that holds files and other directories.

Directory Naming Rules

Rule	Description
Name Length	Up to 255 characters
Characters	Letters, numbers, spaces, and some symbols
Forbidden	/ : * ? " < >
Case Sensitive	Linux is case sensitive; Windows is not

Directory Structure Types

1. **Single-Level Directory** — All files in one directory
2. **Two-Level Directory** — Separate directories for each user
3. **Tree-Structured Directory (Hierarchical)** — Nested directories forming a tree

6.3 Path Names

Type	Description	Example (Windows)	Example (Linux)
Absolute Path	Full path from root	C:\Windows\System32	/home/user/file.txt
Relative Path	Path from current directory	\System32	./Documents/file.txt

7. Programming Languages

7.1 What is a Programming Language?

A programming language is a **formal language** used to write instructions that a computer can understand and execute.

7.2 Levels of Programming Languages

1GL → Machine Language (0s and 1s)
2GL → Assembly Language (Mnemonics)
3GL → High-Level Language (C, C++, Java, Python)
4GL → Query Languages (SQL, Report Generators)

7.3 Machine Language (1GL)

Feature	Description
Definition	Lowest-level programming language
Form	Binary code (0s and 1s)
Execution	Directly understood by CPU
Speed	Fastest
Memory	Very efficient
Portability	Not portable (machine-specific)
Ease	Very difficult to understand

Example

```
10110100 00000001 ; Load 1 into register  
10110000 00000001 ; Add 1 to accumulator
```

```
11110000 00000010 ; Halt
```

Advantages**Disadvantages**

Very fast execution	Hard to write and understand
Uses minimal memory	Easy to make mistakes
Full hardware control	Not portable
—	Writing programs takes long time

7.4 Assembly Language (2GL)

Feature	Description
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Definition	Uses mnemonics to represent machine instructions
Form	Symbols and abbreviations
Execution	Assembler converts to machine code
Speed	Very fast
Memory	Efficient
Portability	Not portable
Ease	Easier than machine language

Example

```
MOV AX, 1 ; Move 1 to register AX
ADD AX, 1 ; Add 1 to AX
HLT      ; Halt
```

Advantages**Disadvantages**

Very fast	Not portable
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Uses minimal memory	Still difficult to write
Full hardware control	Slower than high-level
Easier than machine language	—

7.5 High-Level Language (3GL)

Feature	Description
Definition	Uses English-like statements
Form	Keywords, variables, functions
Execution	Compiler/Interpreter converts to machine code
Speed	Slower than low-level languages
Memory	Less efficient than low-level
Portability	Highly portable
Ease	Easy to learn and use

Examples

Language	Purpose
C	System programming
C++	Object-oriented programming
Java	Cross-platform development
Python	General purpose, AI, data science
JavaScript	Web development
PHP	Web development
C#	.NET development

Ruby	Web development
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Advantages**Disadvantages**

Easy to learn — English-like syntax	Slower than machine/assembly
Portable — runs on different machines	Uses more memory
Faster development	Limited hardware control
Large ecosystem of libraries	—

7.6 Fourth Generation Language (4GL)

Feature**Description**

Definition	Non-procedural, closer to human language
Form	Natural-like statements
Execution	Very high-level
Purpose	Database queries, reports, applications

Examples

Language**Purpose**

SQL	Database queries
Report Generators	Creating reports
Application Generators	Rapid application development
Query Languages	Data retrieval

Advantages**Disadvantages**

Very easy — minimal coding required	Not suitable for all applications
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Very fast development	Limited hardware control
Natural-like syntax	Slower than 3GL
Easy to modify	—

8. Language Translators

8.1 What are Language Translators?

Language translators are **system software** that convert source code written in a programming language into machine code that can be executed by the computer.

8.2 Types of Translators

Type	Description	Input	Output	Speed
Assembler	Converts assembly to machine code	Assembly Code	Machine Code	Fast
Compiler	Converts entire program to object code	High-Level Code	Object Code	Fast
Interpreter	Converts and executes code line by line	High-Level Code	Executes Directly	Slower

8.3 Assembler

What is an Assembler?

An assembler is a program that **converts assembly language code** into machine code.

How Assembler Works

1. Reads assembly code
2. Replaces mnemonics with binary code
3. Allocates memory addresses
4. Generates machine code
5. Creates object file

Features of Assembler

Feature	Description
Macro Support	Handles macros
Symbol Table	Manages symbols and addresses
Error Detection	Detects syntax errors
Output	Produces object code

8.4 Compiler

What is a Compiler?

A compiler is a program that **translates an entire program** written in a high-level language into machine code before execution.

How Compiler Works

Step	Description
1. Lexical Analysis	Checks and groups characters into tokens
2. Syntax Analysis	Checks grammar and structure
3. Semantic Analysis	Checks meaning and logic
4. Code Generation	Generates machine code
5. Code Optimization	Optimizes generated code
6. Object Code Generation	Produces object file

Compiler Process

Source Code → Compiler → Object Code → Linker → Executable File

Features of Compiler

Feature	Description
Speed	Execution is fast
Error Reporting	Reports all errors at once
Optimization	Optimizes code
Portability	Generates machine code for specific platform

Advantages	Disadvantages
Fast execution	Harder to debug
Source code protected	Takes time to compile
Efficient code generation	Generated code is large
No translator needed during execution	—

Examples of Compilers

Compiler	Language	Platform
GCC	C, C++	Linux, Windows
Turbo C++	C, C++	Windows
Java Compiler	Java	Cross-platform
Visual Studio	C#, C++	Windows

8.5 Interpreter

What is an Interpreter?

An interpreter is a program that **converts and executes** high-level code **line by line** during runtime.

How Interpreter Works

1. Reads one line of source code
2. Checks for syntax errors
3. Converts line to machine code
4. Executes the line
5. Moves to next line

Features of Interpreter

Feature	Description
Line-by-Line	Executes code one line at a time
Immediate Execution	Code runs immediately
Error Detection	Stops at first error
Debugging	Easier to debug

Advantages	Disadvantages
Easy debugging — finds errors easily	Slower than compiled code
No compilation — saves compilation time	Source code needed each time
Platform independent	Source code visible
Interactive — good for learning	—

Examples of Interpreters

Interpreter	Language	Platform
Python Interpreter	Python	Cross-platform
PHP Interpreter	PHP	Web servers
JavaScript Engine	JavaScript	Browsers
BASIC Interpreter	BASIC	Various platforms

8.6 Compiler vs Interpreter

Feature	Compiler	Interpreter
Translation	Entire program	Line by line
Execution	After compilation	During translation
Speed	Faster	Slower
Error Reporting	All errors at once	Stops at first error
Debugging	Harder	Easier
Memory	More efficient	Less efficient
Security	Source code hidden	Source code visible
Platform	Platform dependent	Platform independent
Examples	C, C++, Java	Python, PHP, JavaScript

9. Application Software Details

9.1 Word Processing Software

What is Word Processing?

Word processing software is used to **create, edit, format, and print** text-based documents.

Features

Feature	Description
Text Editing	Insert, delete, copy, paste
Formatting	Fonts, sizes, styles
Spell Check	Grammar and spell checking
Templates	Pre-designed documents
Images	Insert and edit images
Tables	Create and format tables
Printing	Print documents

Examples

Software	Developer	Platform
Microsoft Word	Microsoft	Windows, macOS
Google Docs	Google	Web, Android, iOS
LibreOffice Writer	Open Source	Windows, Linux, macOS

Notepad++	Open Source	Windows
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9.2 Spreadsheet Software

What is Spreadsheet Software?

Spreadsheet software is used to **organize, analyze, and calculate** data in rows and columns.

Features

Feature	Description
Cells	Basic unit of spreadsheet
Formulas	Perform calculations
Functions	Built-in calculations (SUM, AVERAGE)
Charts	Visual representation of data
Sorting	Arrange data in order
Filtering	Display specific data
Pivot Tables	Summarize data

Examples

Software	Developer	Platform
Microsoft Excel	Microsoft	Windows, macOS
Google Sheets	Google	Web, Android, iOS
LibreOffice Calc	Open Source	Windows, Linux, macOS

9.3 Presentation Graphics Software

What is Presentation Software?

Presentation software is used to create **visual presentations** with slides, text, images, and animations.

Features

Feature	Description
Slides	Individual pages
Templates	Pre-designed layouts
Animations	Movement effects
Transitions	Slide changes
Multimedia	Insert videos, audio
Presenter View	Notes during presentation

Examples

Software	Developer	Platform
Microsoft PowerPoint	Microsoft	Windows, macOS
Google Slides	Google	Web, Android, iOS
LibreOffice Impress	Open Source	Windows, Linux, macOS
Prezi	Prezi Inc.	Web

10. Comparison Tables

10.1 Types of Software Summary

Type	Description	Examples
System Software	Manages hardware and resources	OS, Utilities, Language Translators
Application Software	Performs specific tasks	Word, Excel, Photoshop
Programming Languages	Used to write programs	C, Java, Python
Translators	Convert code to machine language	Compiler, Interpreter, Assembler

10.2 Operating Systems Comparison

Feature	DOS	Windows	Linux
Developer	Microsoft	Microsoft	Open Source
Interface	Command-line	GUI	GUI + CLI
Multi-User	No	Yes	Yes
Multi-Tasking	No	Yes	Yes
Security	Low	Medium	High
Cost	Free/Commercial	Commercial	Free
File System	FAT	NTFS, FAT	ext4, XFS

10.3 Language Translators Comparison

Feature	Assembler	Compiler	Interpreter
Input	Assembly	High-Level	High-Level
Output	Machine Code	Object Code	Executed Directly
Speed	Fast	Fast	Slow
Error Reporting	Limited	All errors	First error
Debugging	Hard	Hard	Easy

11. Important Full Forms

Abbreviation	Full Form
OS	Operating System
FAT	File Allocation Table
NTFS	New Technology File System
SQL	Structured Query Language
GUI	Graphical User Interface
CLI	Command Line Interface
DOS	Disk Operating System
1GL	First Generation Language
2GL	Second Generation Language
3GL	Third Generation Language
4GL	Fourth Generation Language

Questions

Short Answer Questions (2-5 Marks)

1. What is computer software? What is its need?
2. What is system software? Give examples.
3. What is application software? Give examples.
4. What is an operating system? Name its functions.
5. What is DOS? Write its features.
6. What is FAT? Explain FAT32.
7. What is the difference between FAT32 and NTFS?
8. What is a file naming rule?
9. What are the types of programming languages?
10. What is the difference between compiler and interpreter?
11. What is an assembler?
12. What are the advantages and disadvantages of machine language?
13. What is the difference between high-level and low-level languages?
14. What is 4GL? Give examples.
15. What is the difference between system software and application software?

Long Answer Questions (10 Marks)

1. What is computer software? Explain different types of software in detail.
2. What is an operating system? Explain its functions and types.
3. Explain DOS, Windows, and Linux operating systems in detail.
4. What is a programming language? Explain machine, assembly, high-level, and 4GL languages.
5. Explain the difference between compiler, interpreter, and assembler.
6. What are file allocation tables? Explain FAT, FAT32, and NTFS.
7. Explain file and directory naming rules in detail.
8. What is application software? Explain word processing, spreadsheet, and presentation software.
9. Explain the features and history of Windows OS.
10. What is the difference between FAT32 and NTFS file systems?

Objective Questions (1 Mark)

1. Which is system software? (a) MS Word (b) Windows (c) Excel (d) PowerPoint

Answer: (b)

2. Which is not an operating system? (a) Windows (b) Linux (c) MS DOS (d) MS Word

Answer: (d)

3. The brain of computer is: (a) Memory (b) CPU (c) Hard Disk (d) Software

Answer: (b)

4. Assembly language uses: (a) 0s and 1s (b) Mnemonics (c) English-like words (d) None

Answer: (b)

5. Compiler translates: (a) Assembly to machine (b) High-level to machine (c) Both (d) None

Answer: (b)

6. Interpreter executes code: (a) Entire program (b) Line by line (c) Both (d) None

Answer: (b)

7. FAT32 maximum file size is: (a) 2 GB (b) 4 GB (c) 8 GB (d) 16 GB

Answer: (b)

8. Which is not a language translator? (a) Compiler (b) Interpreter (c) Assembler (d) Editor

Answer: (d)

End of Unit-IV Notes